Computers, Environment and Urban Systems 82 (2020) 101501

Contents lists available at ScienceDirect

Computers, Environment and Urban Systems

journal homepage: www.elsevier.com/locate/ceus

Scale effects in remotely sensed greenspace metrics and how to mitigate them for environmental health exposure assessment

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ARTICLE INFO

Keywords:
Urban
Greenness exposure
Health
GIS
Buffer distance
Lacunarity

ABSTRACT

Metrics representing exposure to the natural environment are widely used in environmental health-related studies. They are calculated using a variety of different data sources representing greenspace and a range of buffer sizes representing human interaction with the environment. Previous studies have identified issues relating to buffer distance and scaling effects on greenspace exposure assessments when using satellite image-derived metrics. We evaluate the spatial scale sensitivity of three common greenspace metrics (i.e., Normalised Difference Vegetation Index- NDVI, Leaf Area Index- LAI, and Land Use and Land Cover-LULC), using lacunarity analysis, as a scale-dependent measure of heterogeneity based on the principles of fractals. By producing a 'lacunarity curve' across multiple spatial scales, we defined the scale-variances for specific greenspace metrics, including the upper scale limit at which the metrics become invariant, approximately 640 m for Sentinel-2 and 480 m for Landsat-8. Each of the greenspace metrics we considered exhibited scale sensitivities, meaning that each is expected to have a different influence on the strength and significance of the statistical associations found between greenspace exposure and health depending on the spatial scale of analysis (e.g., buffer distance). Using lacunarity curves, we produced a novel composite, multi-scale greenspace 'exposure index' in which each input scale is weighted according to its relative scale sensitivity. We also created a multi-scale, multi-metric map combining the different vegetation measures while accounting for scale. We found that cumulative exposure gradients across a large urban conurbation are even more marked when using our multi-scale 'exposure index' maps compared to traditional approaches. Our multi-scale, composite greenspace 'exposure index' mapping techniques are not as vulnerable to scale effects as traditional approaches and can be readily transferred to the analysis of other environmental exposure variables such as air pollution.

1. Introduction

Emerging scholarly literature has, in the past decade, focused on the direct and indirect benefits of the urban natural environment on human health and wellbeing (Hartig, Mitchell, De Vries, & Frumkin, 2014; James, Banay, Hart, & Laden, 2015; Kondo, Fluehr, McKeon, & Branas, 2018; Nieuwenhuijsen, Khreis, Triguero-Mas, Gascon, & Dadvand, 2017). Multiple original studies, as well as scholarly reviews of the extant literature, have indicated that green or blue space (hereafter greenspace) exposure generally produces net positive benefits on health (Gascon et al., 2015; Helbich, De Beurs, Kwan, O'Connor, & Groenewegen, 2018; Lindley, Cook, Dennis, & Gilchrist, 2019; Rojas-Rueda, Nieuwenhuijsen, Gascon, Perez-Leon, & Mudu, 2019; Nieuwenhuijsen et al., 2019). These benefits accrue both from the therapeutic qualities of greenspace and its inherent ability to provide essential ecosystem functions related to human survival (Bratman et al., 2019; Costanza et al., 2017; Kabisch, Qureshi, & Haase, 2015; Lindley, Pauleit, Yeshitela, Cilliers, & Shackleton, 2018).

It must be noted, however, that some researchers have recently found insignificant and negative or mixed associations between green-blue space exposure and health indicators (Ali, Di Nardo, Harrison, & Verma, 2017; Fuertes et al., 2014; Kondo et al., 2018; Markevych et al., 2016; Picavet et al., 2016). Labib, Lindley, and Huck (2020) and Markevych et al. (2017) argued that such conflicting results and inconsistency require more specific and process-oriented research to gain a better understanding of the underlying complexities between greenspace exposure and health. In line with this argument, several recent studies have investigated the impact of differences in methodological

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https://doi.org/10.1016/j.compenvurbsys.2020.101501
Received 19 December 2019; Received in revised form 18 March 2020; Accepted 3 May 2020
Available online 21 May 2020
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approaches on quantifying and representing greenspace exposure linking the effects of such exposure to health outcomes. These studies focused on understanding the influence of spatial scale (in terms of both the size and shape of measurement units) on the relationship between greenspace exposure and health indicators (Su, Dadvand, Nieuwenhuijsen, Bartoll, & Jerrett, 2019, Zhang and Tan, 2019; Cebrecos, Domínguez-Berjón, Duque, Franco, & Escobar, 2018; Tsai, Leung, McHale, Floyd, & Reich, 2018). They indicated that spatial scale and the Modifiable Areal Unit Problem (MAUP) (Openshaw, 1981) influence the reported associations found between greenspace exposure and health. As such, investigations into the impact of these scale-related factors and the development of alternative approaches that are less sensitive to these issues are critical to understanding the impact of the association between exposure and health (Labib et al., 2020).

Measuring greenspace exposure by creating a discrete neighbourhood zone around specific receptor points (e.g. home addresses) is one of the most widely applied methodological approaches in studies investigating the relationship between greenspace and health (Browning & Lee, 2017; James et al., 2014; James, Hart, Banay, & Laden, 2016). With the wide availability of satellite images (e.g., Landsat, Sentinel-2), an increasing number of studies use buffer analysis (a method of zoning around map features based on distance or estimated travel time) to derive metrics from satellite imagery representing relative levels of 'greenness' (James et al., 2015; Markevych et al., 2017; Su et al., 2019). However, there is no consensus among scholars as to what buffer distance should be used to define the exposure area (e.g., neighbourhood). Indeed, it is highly unlikely that such a universal figure could ever be justified (Huck et al., 2018a; Labib et al., 2020). Additionally, there are several metrics used as the basis to measure greenspace exposure, including the Normalised Difference Vegetation Index (NDVI), Land Use and Land Cover (LULC), canopy cover (CC) and Leaf Area Index (LAI), all of which are proxy measures and so can produce different results for a given location and buffer distance (Browning & Lee, 2017; Labib et al., 2020; Reid, Kubzansky, Li, Shmool, & Clougherty, 2018; Su et al., 2019).

Su et al. (2019), Zhang and Tan, 2019, Reid et al. (2018), and Browning and Lee (2017) demonstrated that variation in both the buffer distances used to define the unit of analysis and the spatial resolution of satellite images are susceptible to the MAUP and may therefore produce scale effects in greenspace exposure and health studies (see also Houston, 2014). Although these studies highlight issues relating to varying spatial scales and their relationship to greenspace metrics and the MAUP, they do not adequately investigate the ways in which scale poses an issue for such analyses, nor how multiple scales and exposure metrics can be comprehensively measured to overcome or reduce scale effects.

Previous studies have focused on measuring greenspace exposure in terms of specific individuals' home (usually within the neighbourhood), school, or office addresses. However, this approach limits the possibility of taking account of greenspace exposure beyond the given neighbourhood or defined exposure area (Frumkin et al., 2017; Helbich, 2018; Pearce, 2018; Zhang, Zhou, Kwan, Chen, & Lin, 2018) and eliminates the possibility of holistically gaining an understanding of the spatial variability of exposure for the whole urban landscape, where people work, live, and interact with nature (Helbich, 2018; Helbich, 2019a; Park & Kwan, 2017; Zhang et al., 2018). As such, studies utilising such approaches cannot provide a comprehensive understanding of the greenspace exposure experienced within a larger urban area (Frumkin et al., 2017; Labib et al., 2020; Zhang et al., 2018). We propose a new greenspace exposure mapping methodology to create a map of greenspace exposure for the entire urban area under consideration, rather than conducting individual exposure assessments at a finite number of fixed addresses and represent exposure gradients across entire case study areas without the need for separate buffer analysis.

The objectives of this study are to:

(i) investigate how different satellite-derived measures of 'greenness' are impacted by spatial scale variations using lacunarity analysis, (ii) create greenspace 'exposure' maps at specific scales (e.g., where the scale of exposure pathways for particular health outcomes might be known) and generate multi-scale composite maps using the lacunarity results (e.g., where exposure pathways and their scales are less well known), (iii) propose an improved multi-scale, multi-metric greenspace 'exposure index' map. The scale-composite maps not only take into account scale effects and improve representativeness but can also improve the efficiency of greenspace exposure assessments. We also assess the implications of the new method by investigating how exposure profiles and correlations with social and health deprivation changed for a large urban case study area.

2. Background

2.1. Greenspace exposure and greenspace metrics

Zartarian, Bahadori, and McKone (2005) state that exposure assessment is *"the process of estimating or measuring the magnitude, frequency, and duration of exposure to an agent, along with the number and characteristics of the population exposed. Ideally, it describes the sources, pathways, routes, and the uncertainties in the assessment"* (p3). For greenspace exposure, it implies exposure to nature/natural environments (e.g., gardens, stands of trees, parks etc.) and the magnitude, frequency and duration of its impact on humans (Bratman et al., 2019; Frumkin et al., 2017). From the existing literature, from a spatial perspective greenspace exposure can primarily be categorised as (i) availability exposure (the magnitude of greenness available around the study subject), (ii) accessibility exposure (the accessibility to and from designated greenspace, e.g., from a park to a home or work address), (iii) visibility exposure (the amount of greenspace that is visible to the study subject) (Dadvand & Nieuwenhuijsen, 2019; Dzhambov, Markevych, & Lercher, 2019; Labib et al., 2020). These categories of greenspace exposure often overlap whilst having cumulative effects on human health. The present study only focuses on the exemplar of availability exposure due to its dominance in previous studies. Several types of greenspace metrics are current in use to measure availability exposure and they broadly fall into three groups; (a) remote sensing-based indices (e.g., NDVI, Enhanced Vegetation Index, LAI); (b) percentage of green cover calculated using a LULC dataset; (c) numbers of trees or canopy coverage (Gascon et al., 2016; Labib et al., 2020; Orioli et al., 2019; Dadvand et al., 2017).

2.2. Scale, MAUP, and uncertainty in greenspace exposure assessment

In this study spatial scale is defined as, "*the degree of spatial reduction indicating the length used to represent a larger unit of measure; ratio of distance on the map to distance on the earth surface represented by the map*" (Turner & Gardner, 2015, p 18). According to this definition, large scale represents a finer resolution (or smaller buffer distances), and small scale represents a coarser resolution (or larger buffer distances). For greenspace exposure, different buffer distances produce the exposure areas (e.g., neighbourhoods) within which greenspace exposures are measured, and therefore generate varying scales of analysis for the influences of greenspace exposure upon health (Zhang and Tan, 2019; Labib et al., 2020). The impact of spatial scale on measuring a variable and statistical inference is widely established, and is known as the MAUP (Jelinski & Wu, 1996; Openshaw, 1981). The MAUP involves two interrelated issues: (i) a zoning effect, representing the variance in results because of alternative areal unit delineations; and (ii) a scale effect, reflecting the sensitivity of the results and inferences to areal unit size (Dark & Bram, 2007; Houston, 2014).

In past studies of the relationship of greenspace to health, the MAUP usually occurs as the greenspace exposure measurements are aggregated to different buffer sizes (Zhang and Tan, 2019; Houston, 2014). Houston (2014) demonstrated that both the zoning and the scale

effect are present due to aggregation over different buffer sizes and the use of arbitrary zones. Zhang and Tan (2019) argued that the relationship between greenspace and health is poor or insignificant at smaller spatial scales (coarser resolutions) such as the city scale, but at larger spatial scales (finer resolutions) the results are inconsistent. Conversely, Su et al. (2019) demonstrated that relatively large buffer sizes (e.g., 500 m) are significantly associated with physical health outcomes, though the effects upon mental health and other perceived health outcomes diminish below the 250 m distance, thus the scale effects on the relationships vary over different buffer distances used to set the spatial scales. The issue of inconsistency in effect size as buffer distance increases (e.g., usually beyond 500 m) is identified in several other studies (Browning, Lee, & Wolf, 2019; Dadvand et al., 2014; Engemann et al., 2019; Orioli et al., 2019; Wolch et al., 2011). These studies indicate variation in effect size and significance on the association between greenspace exposure and health according to the scale of analysis. We contend that such a phenomenon is most likely related to the presence of scale effects.

Due to a lack of widely-recognized theoretical guidelines regarding what buffer distance is the optimum for which type of study, most recent studies have utilised multiple buffers at varying scales and explored the sensitivity of health indicators to the different buffer distances in a seemingly arbitrary fashion (Browning & Lee, 2017; Labib et al., 2020). Markevych et al. (2017) argued that the selection of buffer size and areal unit of analysis might depend on the pathways of benefits being considered in a given study; for example, for air and noise pollution mitigation, a smaller buffer ($\leq$ 100 m) may be more appropriate, whereas for physical activity based benefits a larger buffer (e.g., 500 m) size might be relevant (Labib et al., 2020). Therefore, it appears that selection of an absolute standard in terms of buffer size or spatial unit of greenspace exposure measurement may not be possible due to how the different potential benefit pathways from greenspace may vary over different spatial extents. However, generic measures may still be important when general health or unknown pathways are being considered.

The issue of the potential benefit pathways from greenspace varying over different spatial extents also relates to the uncertain geographic context problem (UGCoP) identified by Kwan (2012). From a greenspace exposure perspective, the UGCoP indicates that there is spatial uncertainty in the actual areas over which the greenspace exposure can be beneficial for health due to different pathways and contexts (James et al., 2014; Markevych et al., 2017; Pearce, 2018). As a result of UGCoP, it may not be possible to be certain about the scales at which the benefits of greenspace exposure have the greatest impact upon individuals due to their inherent heterogeneity. However, it might be possible to combine multiple scales to overcome some of the issues of spatial uncertainty. By combining several scales related to different pathways of greenspace exposure benefits, it might be possible to produce and map an overall greenspace 'exposure metric' that considers multiple pathways. The current literature, however, has not yet provided any examination or analysis of the consequences of combining different scales and metrics to improve the exposure measurement for overall greenspace benefits along multiple pathways (e.g., temperature and air pollution regulation or physical activities).

2.3. Lacunarity: concept and application to satellite images

Dark and Bram (2007) and Tate (2001) recommend several approaches to understand and reduce the scale and zoning effects in pixel (raster cell) data, including the analysis of fractal dimensions. Previous studies have shown that satellite images and related data illustrate complex characteristics and scale effects due to the possible presence of fractal or multifractal dimensions (see Mandelbrot, 1967; Mandelbrot, 1983; Myint & Lam, 2005) that can be further explained by lacunarity (Ariza-Villaverde, Jiménez-Hornero, & De Ravé, 2013; Malhi & Román-Cuesta, 2008).

Mandelbrot (1994, 1983) introduced the concept of lacunarity (a measure of the textural properties of an image) in order to distinguish texture patterns that have the same fractal dimension but are actually dissimilar in appearance. Lacunarity describes the distribution of gap sizes in a texture that may or may not have fractal dimensions (Malhi & Román-Cuesta, 2008; Myint & Lam, 2005). Gefen, Meir, Mandelbrot, and Aharony (1983) defined lacunarity as a measure of the deviation of a geometric object, such as a fractal, from translational invariance. For imagery, lacunarity therefore measures the spatial heterogeneity of objects depending on the texture of the image at varying scales (Ariza-Villaverde et al., 2013; Myint & Lam, 2005). Consequently, an image with a higher number of gaps has higher lacunarity versus an image with a smaller number of gaps (or transitional invariance) (Myint & Lam, 2005). However, the 'gappiness' or heterogeneity of the image texture is dependent on the scale of the measurement, as objects that are heterogeneous at large scales can be homogeneous at small scale or vice versa (Malhi & Román-Cuesta, 2008; Plotnick, Gardner, Hargrove, Prestegaard, & Perlmutter, 1996).

In the case of greenspace exposure, as the spatial scale and spatial unit of analysis changes, so does the spatial heterogeneity of objects in satellite images (or derived indices e.g., NDVI) (Malhi & Román-Cuesta, 2008; Myint & Lam, 2005; Tate, 2001). This relationship between the scale of analysis and spatial heterogeneity of the dataset is of great practical importance, as increasing the buffer size beyond a certain level will mean that the sample of the landscape that falls within the buffer will become increasingly homogeneous as the size of the buffer surpasses that of the repeating patterns in the landscape (e.g., tree cover). Similarly, increasing the cell size in the dataset (reducing the spatial resolution) would lead to the aggregation of smaller features in the landscape, again leading to increasing homogeneity in a sample taken from the dataset. In both cases, this homogeneity will mask variations in exposure to greenspace, meaning that the results of any analysis are highly dependent upon the relationship between the chosen scale and spatial unit. Based upon this premise, further analysis of the greenspace metrics commonly used in greenspace exposure assessments was performed. This analysis seeks to explain why scale is an issue for greenspace exposure assessment and to explore new approaches to the combination of multi-scale measurements into a single greenspace 'exposure index' to overcome these issues.

3. Methods and materials

3.1. Case study area

The methodological processes developed for the present study were applied as a case study to Greater Manchester Metropolitan County, situated in the North West of England. Greater Manchester is a post-industrial city-region comprising an area of about 1276 km^2 and a population of 2.8 million people (Dennis et al., 2018). Physically, the city-region possesses a diverse landscape pattern, with flat plain areas that give way to hills that rise to 500 m in the north and east. These uplands in the north and east contain the headwater sources of several rivers (e.g., River Mersey) that flow throughout the city-region and beyond (Dennis et al., 2018). From the perspective of greenspace studies the Greater Manchester area is of particular interest due to its diverse landscape pattern, availability of various greenspace elements (e.g., including urban parks of various sizes, green-belts, and pastoral farm land), and the Greater Manchester Combined Authority's strategies to develop the city-region in a way that promotes sustainability, green infrastructure and biodiversity (Labib, 2019).

3.2. Greenspace metrics

To identify the underlying mechanisms of scale-dependent variation on greenspace exposure measurement and in order to prepare a multi-scale greenspace 'exposure index', this study utilised three different

greenspace exposure metrics commonly used in the existing literature, (i) NDVI, (ii) LAI, and (iii) Land Use and Land Cover (LULC) (Labib et al., 2020; Markevych et al., 2017). All three metrics for the main analysis were derived from Sentinel-2A satellite imagery. Sentinel-2 images were selected because they have better spatial resolution (10 m) than the widely used Landsat images (30 m). Additionally, previous studies have noted that Sentinel-2 images yield results with higher overall accuracy for detecting vegetation in urban settings compared to relatively low-resolution imagery such as Landsat-8 (Labib & Harris, 2018; Markevych et al., 2017). In order to better understand the scale sensitivity of Landsat images, we also experimented with NDVI derived from Landsat images (see *Supplementary note*-1).

For the purposes of this study, NDVI and LAI indices were processed from Sentinel-2A images collected on 4th July 2018, with cloud cover < 2% and atmospheric correction was undertaken prior to analysis. NDVI was calculated using the following standard equation:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$

where NIR refers to the near-infrared band (central wavelength 832.8 nm) and RED refers to the red band in the visible region (central wavelength 664.6 nm) (Drusch et al., 2012). The estimated NDVI values ranges between −1 and 1, with higher positive values indicating greater green vegetation cover (Gascon et al., 2016; Su et al., 2019). The LULC based greenspace metric used in this study was obtained from Dennis et al. (2018), who applied a supervised classification algorithm to produce a 10 m spatial resolution dataset of five land cover types; urban, water, tree canopy, grass, forbs and shrubs.

Leaf Area Index (LAI) is an important biophysical variable influenced by levels of photosynthesis, transpiration, and overall energy balance exhibited by the vegetation in the satellite images, all of which are key parameters critical to the provision of beneficial ecosystem functions such as air pollution reduction and cooling functions (Lin & Lin, 2010; Nowak, Hirabayashi, Bodine, & Greenfield, 2014). As a result, LAI values are important metrics contributing to a better understanding of regulating ecosystem functions of vegetation (Elmqvist et al., 2015; Orioli et al., 2019). An LAI map for Greater Manchester was derived using the same Sentinel-2A dataset used for NDVI. The LAI dataset was calculated utilising the L2B biophysical processor toolbox, a component of the open-access software Sentinel Application Platform (SNAP, version 5.2) developed by the European Space Agency (Upreti et al., 2019; Weiss & Baret, 2016). Details of this the toolbox and LAI measurement model can be found in Weiss and Baret (2016).

3.3. Lacunarity analysis

3.3.1. Calculation algorithm

Several algorithms exist to compute lacunarity (e.g., Allain & Cloitre, 1991; Gefen et al., 1983), though the "*gliding box*" algorithm developed by Allain and Cloitre (1991) has received the greatest attention due to its conceptually straightforward and computationally simple approach (Malhi & Román-Cuesta, 2008; Plotnick, Gardner, & O'Neill, 1993; Roces-Díaz, Díaz-Varela, Álvarez-Álvarez, Recondo, & Díaz-Varela, 2015). Plotnick et al. (1993, 1996) popularised this algorithm and illustrated lacunarity estimation to not only be applicable for fractal dimensions, but also applicable to understanding landscape patterns and scale variations. This algorithm has therefore been used to estimate lacunarity in this study.

The algorithm workflow commences with a square box neighbourhood (similar to neighbourhood representations through circular buffer zones) of size '$r \times r$' (e.g., 2×2) which is placed at one corner of the raster map (e.g., satellite image). This box in turn determines the spatial unit of measurement (i.e., the scale of analysis), and the 'box mass' $S(r)$, is computed for the pixels that fall within the box using an algebraic function (i.e., sum, range) on pixel values, dependent on the type of data. Lacunarity analysis can be applied to both binary (using sum for the box mass) and continuous images (using range for the box mass) (Dong, 2000; Hoechstetter, Walz, & Thinh, 2011; Malhi & Román-Cuesta, 2008; Plotnick et al., 1993).

Next, the box representing the neighbourhood is shifted one pixel (to the right) and box mass is calculated for the new position. This process is repeated by sliding this box over the whole map M ($M = W \times L$), for an irregular shape map W denotes the width of the map, and L denotes the length. For the M size map, the total number of boxes of size r is $N[r]$ (Eq. 2).

$$N[r] = (W - r + 1)X(L - r + 1) \quad (2)$$

After estimating $N[r]$, in the next stage the number of boxes of size r containing box mass S are counted as $n[S,r]$, and this frequency distribution is converted into a probability $Q(S,r)$ distribution by dividing by $N[r]$ (Eq. 3).

$$Q(S,r) = \frac{n[S,r]}{N[r]} \quad (3)$$

Following the above calculation, the first and second moments of this probability distribution are estimated using Eq. (4) and (5) respectively:

$$Z(1) = \sum S \times Q(S,r) \quad (4)$$

$$Z(2) = \sum S^2 \times Q(S,r) \quad (5)$$

Finally, these moments of the lacunarity $\Lambda(r)$ of the box size r are computed as (Eq. 6):

$$\Lambda(r) = \frac{Z(2)}{Z^2(1)} \quad (6)$$

Plotnick et al. (1993) recognized that based on the statistical behaviour $\Lambda(r)$ the first and second moment can be expressed as a mean $E[S(r)]$ and variance $var[S(r)]$ of the box mass values for box size r accordingly. Therefore, lacunarity $\Lambda(r)$ can also be computed as (Eq. 7):

$$\Lambda(r) = 1 + \frac{Var[S(r)]}{E^2[S(r)]} \quad (7)$$

3.3.2. Lacunarity of greenspace metrics

In this study, we estimated the lacunarity values for the greenspace metrics at multiple scales using a five-step method. The overall work procedure is illustrated in Fig. 1. The details of these steps are as follows. We used ArcGIS 10.4 (ESRI, US) and MS Excel for lacunarity analysis.

3.3.2.1. Step 1: image processing. The first step (Fig. 1, Step 1) involves determining whether the input image should be interpreted as binary or continuous. Categorical data in the LULC map were converted to a binary map layer where 'Canopy', 'Grass', and 'Forbs and Shrubs' classes were reclassified as green (value of 1) and other classes as not-green (value of 0). LAI and NDVI data were handled as continuous data and were normalised to the same value range. While both datasets might have been handled as categorical data, this would have involved introducing an artificial, arbitrary boundary between values, negatively affecting the lacunarity results.

3.3.2.2. Step 2: box mass estimation. In the second step, (Fig. 1), box mass was estimated for box size r, for each binary and continuous map. For the LULC binary image Focal Statistics 'Sum' was selected and for NDVI and LAI 'Range' was utilised on the basis of the algorithm discussed in Section 3.3.1. For each greenspace metric, lacunarity was calculated for box sizes as functions of $2^p + 1$, where p is a whole number commencing at 1 and increasing to a limit defined by the box size being half of the shorter dimension of the study region (map width or height) (Huck et al., 2018b; Sui & Wu, 2006). For the purposes of the present study, we started from $r = 3$ (*where, $p = 1$*), and went up to

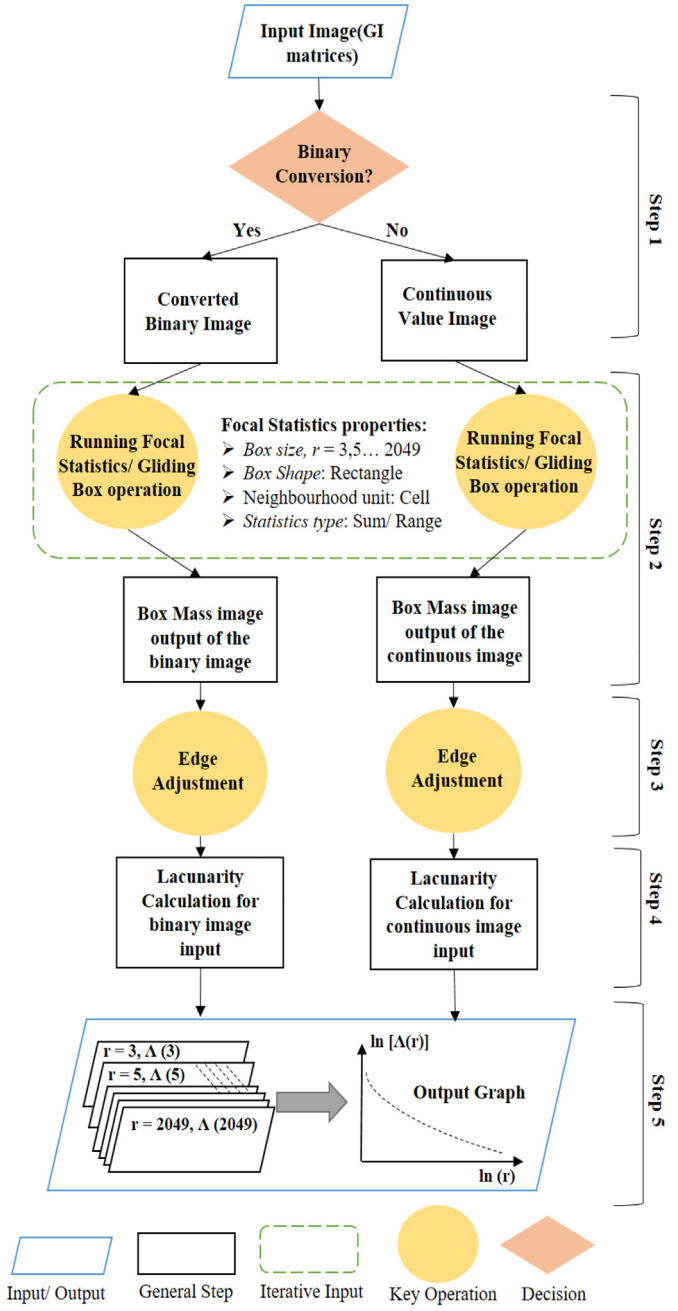


Fig. 1. Flow diagram for estimation of lacunarity for different greenspace metrics and presentation of results in relation to different scales.

$r = 2049$ (*where, p = 11*) at which point the size of the gliding box became half of the width of the study area.

3.3.2.3. Step 3: edge adjustment. The irregular shape of the study area (being neither a square nor a rectangle) means that the presence of 'No Data' pixels outside the study area effectively reduces the size of the box at the study area boundary. Step 3 (Fig. 1) corrects for these edge effects using the procedure identified in Sui and Wu (2006). Accordingly, a weighting factor was applied to adjust the box mass of the gliding boxes containing areas including 'No Data' pixels (see *Supplementary note*-2).

3.3.2.4. Step 4: calculation of lacunarity. After edge adjustment, at Step 4 (Fig. 1) we estimated the lacunarity value for each box size for all three datasets. In the case of the binary LULC map we applied Eq. (2)–(6) (Section 3.3.1), and for the NDVI and LAI we used Eq. (7) to estimate lacunarity for the box sizes we considered (for each *r*).

3.3.2.5. Step 5: representing scale dependent lacunarity. At Step 5 (Fig. 1), we calculated the relationship between scale (gliding box size) and lacunarity for each greenspace metric. As discussed earlier, lacunarity is a scale dependent variable, and the most meaningful information is obtained by plotting lacunarity values against their respective different box sizes in order to create a lacunarity curve. The most widely established procedural method is to plot the natural logarithm (ln) of both lacunarity (Y-axis) and box size (X-axis) in a *log-log* plot (Hoechstetter et al., 2011; Malhi & Román-Cuesta, 2008; Roces-Díaz et al., 2015).

3.4. Scale-specific and multi-scale composite mapping

After estimating lacunarity, we used the results to identify the upper scale limit beyond which the measurement of each greenspace metric became invariant due to landscape homogeneity due to aggregation. Then, we prepared greenspace 'exposure index' maps for each metric for the whole study area using multiple specific scales below the defined upper limit. Here we use the term 'exposure index' map to mean areas of potential exposure should a receptor be located in each grid cell of the map. For each specific scale, mean values of NDVI, LAI, and green percentage (for LULC) were estimated. Finally, a single multi-scale 'exposure' map for each greenspace metric was created using a weighted average method based on weights obtained from the lacunarity analysis (Table 1). Lacunarity values were used as weights due to their objective representation of the relative spatial heterogeneity and scale-sensitivity of each of the metrics. Combined multi-scale 'exposure' maps for each greenspace metric were calculated using (Eq. 8).

$$CGS_m = \frac{\sum_{i=1}^{L} (GS_i \times W_i)}{\sum_{i=1}^{L} W_i} \quad (8)$$

In (eq8), CGS_m is the combined multi-scale 'exposure' value for each greenspace metric m (m = NDVI, LAI, LULC); GS_i is the individual value for the metric at scale i; and W_i is the weight of the scale i (i can be 1 to L numbers of scales, in our case $L = 5$). This procedure produced three multi-scale greenspace 'exposure index' maps (i.e., CGS_{NDVI}, CGS_{LAI}, and CGS_{LULC}) for our study area.

Finally, the metrics were combined into a single result, completing the third objective of the study. The resultant multi-scale, multi-metric composite greenspace 'exposure index' map was prepared from the three multi-scale composite maps for each metric. The CGS_{NDVI}, CGS_{LAI}, and CGS_{LULC} were reclassified into nine categories using a natural breaks classification method (Jenks, 1977), with the lowest 'exposure' range reclassified as 1 and the highest range as 9 (though a continuous standardised scale could alternatively be used). The three reclassified maps were then combined using (Eq. 9) to produce a composite multi-scale, multi-metric, greenspace 'exposure index' map.

$$GS_{in} = \frac{(CGS_{NDVIr} + CGS_{LAIr} + CGS_{LULCr})}{T} \quad (9)$$

In (Eq. 9), GS_{in} is the composite greenspace 'exposure index' value,

Table 1
Scales, box size and weights for the five scales considered in the multi-scale composite maps.

Sl. scales [*i*]	Box size (in metres)	Weighting- [w] (Lacunarity value)			
		Wi	*NDVI*	*LAI*	*LULC*
1	100	1	0.070	0.201	0.671
2	200	2	0.036	0.111	0.655
3	300	3	0.027	0.081	0.646
4	400	4	0.023	0.067	0.638
5	500	5	0.024	0.060	0.631

and CGS_{NDVIr}, CGS_{LAIr}, and CGS_{LULCr} are reclassified multi-scale greenspace metric 'exposure' value. T is the number of metrics (in our case, $T = 3$). To represent the composite greenspace 'exposure' level, we classified the GS_{in} map into nine classes and ranked composite GS_{in} values from lowest to highest. A GS_{in} value of 1 indicated *Extremely Low* and 9 represented *Extremely High* levels of greenspace 'exposure'. All outputs were masked to exclude the extra cells generated outside the study area boundary.

3.5. Assessment of the composite greenspace exposure maps

To understand the potential implications of our outputs, we assessed how greenspace exposure profiles differ when using our new multi-scale (and multi-scale, multi-metric) composite greenspace 'exposure index' maps compared with traditional measures. The comparison was carried out by examining deciles of social deprivation, identified from the index of multiple deprivation (IMD) and health deprivation (English Indices of Deprivation, 2015) for Greater Manchester neighbourhoods based on Lower Super Output Areas (LSOA). LSOAs have a mean population of around 1500 and range from 1000 to 3000 people (Daras, Green, Davies, Barr, & Singleton, 2019). We considered IMD and health deprivation deciles as indicators of socioeconomic deprivation and pseudo health conditions for neighbourhoods in Greater Manchester (Cebrecos et al., 2018; Mitchell & Popham, 2008). Comparisons were conducted in terms of the variations in mean values for composite greenspace exposure, percentage of greenspace, and NDVI for different IMD deciles (where the lowest deciles indicate the highest deprivation). Additionally, we used Spearman's rank correlation to assess correlations between the IMD, and health deprivation metrics and our composite exposure metric and traditional metrics.

4. Results

4.1. The greenspace metrics for case study area

Fig. 2 shows that the distribution of greenspace had a common pattern among all three metrics. Large areas of vegetation were found at the edge of the study area (Fig. 2b, d), and there were several green corridors. However, the LAI values (Fig. 2c) indicated fragmentation of vegetation cover and thick vegetative cover was scarce in the study area. The LULC map (Fig. 2d) shows several concentrations of built-up areas, which correspond to the urban core areas of the ten local authorities (Fig. 2a). Overall, the study area possessed a moderate spread of vegetative cover of various types (e.g., grass, trees), with green areas surrounding the urban core areas as well as fragmented areas of vegetation within the urban core.

4.2. Lacunarity and scale effect on greenspace metrics

The lacunarity analysis results for all greenspace metrics are summarised in Fig. 3. *Log*-log lacunarity curves displayed in Fig. 3(a) clearly indicate that changes in box size, i.e., changes in the spatial scale of analysis or buffer distances, affects lacunarity values across all greenspace metrics. Thus, the scale dependency of the greenspace metrics was confirmed. As illustrated in Fig. 3(a), at smaller spatial units of analysis lacunarity values were the highest, and as the box size gradually increased (i.e., the areal unit increased), the ln lacunarity values tended to be close to zero. This implies that there is an inverse relationship between the increasing size of spatial units (i.e., measurement box size) and the resulting variance in each greenspace metric.

Changes in lacunarity can be better understood by observing the three rows of images for each metric presented in Fig. 3. From Fig. 3b, it is clear that at smaller box sizes ($r = 5$), the variation within the box mass is high, whereas for the value of $r = 33$ the variation within the box mass is reduced, and at $r = 129$ the box mass values show low variation. Finally, at $r = 513$, the box masses become near invariant for the NDVI metric. Thus, with increasing box sizes the variation of box masses tends to zero, which results in lower values of lacunarity. This pattern of variance in box mass is also similar for both LAI (Fig. 3c) and LULC (Fig. 3d). In addition to variation in box mass, the variation in the heterogeneity of the landscape characteristics is also visible from Fig. 3 b-d. As the box/buffer size increases the level of detail decreases, thus the landscape that appears heterogeneous at a smaller box size becomes homogenous in appearance at a larger box size due to aggregation effects. As a result, the actual characteristics of the landscape are no longer captured at larger box sizes (smaller scales) because of generalisation of landscape characteristics (e.g., the degree of contagion and the presence of self-similarity; Plotnick et al., 1993; Turner & Gardner, 2015; Hay, Marceau, Dube, & Bouchard, 2001). Scale sensitivity is therefore a critical factor in performing measurements of greenspace exposure studies.

In line with scale sensitivity, the upper scale limit after which variation in the metrics reduced to nearly zero due to further increases in scale can be clearly identified in the lacunarity curve (Fig. 3). In Fig. 3a, it is clear that within the range 300–600 m, the slope of the lacunarity curves for NDVI and LAI becomes flattened. This range can be interpreted as the scale where characteristics of the landscape begin translational homogeneity (Hoechstetter et al., 2011; Malhi & Román-Cuesta, 2008; Roces-Díaz et al., 2015). The range indicates the values after which an increase of buffer size or decrease of scale results in invariance in the greenspace metric measurements. This means that the metrics no longer capture the characteristics of the dominant object (e.g., vegetation), and are merged into a generalised wider landscape (Dale, 2000; Hay et al., 2001; Plotnick et al., 1993). Applying spatial units of analysis beyond this scale range will therefore not result in any substantive impact upon the analysis (e.g., statistical inference). In the case of NDVI and LAI metrics from Sentinel-2 images (10 m resolution), buffer distances beyond ~640 m will produce biased estimations greenspace exposure values due to aggregation of information at scales larger than the landscape characteristics, hence ~640 m can be considered as the upper limit of the acceptable range of scales for Sentinel-2 images. However, for Landsat-8 (30 m resolution) images, we found that the lacunarity curves flatten at ~480 m (see *Supplementary note-1*). This finding suggests that as one might expect, the upper scale limits are smaller for images of lower spatial resolution than for those with relatively high resolutions.

Fig. 3(a) indicates that LULC had the highest range of lacunarity values over several scales, whereas NDVI had the lowest range of values. These values were observed due to the types of data, box mass estimations, and the geometry of the map of each metric. For example, sparse maps usually exhibit higher lacunarity than dense maps (Hoechstetter et al., 2011). Here, the LULC maps had the sparsest geometric pattern (Fig. 2d), and hence the highest lacunarity values. By contrast, NDVI and normalised LAI maps have the densest patterns; as a result, they exhibit lower lacunarity values.

4.3. Greenspace 'exposure index' mapping

In our analysis, the upper limit of the acceptable range of scales for the Sentinel-2 NDVI data was found to be around 640 m. Therefore, for greenspace 'exposure index' mapping, we only selected scales below 640 m. These scales were set by buffer distances of 100, 200, 300, 400, and 500 m. This range of scales supports the multiple pathways of benefits of greenspace exposure. Buffer distances such as 100 and 200 m, for example, allow consideration of relatively localised exposure benefits, e.g. some forms of air pollution removal and cooling (Labib et al., 2020; Markevych et al., 2017); whereas distances such as 300, 400, and 500 m allow the consideration of neighbourhood exposure benefits in terms of the capacity-building benefits of greenspace exposure such as encouraging physical activities, improving social cohesion, and stress recovery (Hartig et al., 2014; Markevych et al., 2017; Nieuwenhuijsen et al., 2017). Fig. 4 illustrates the NDVI exposure maps

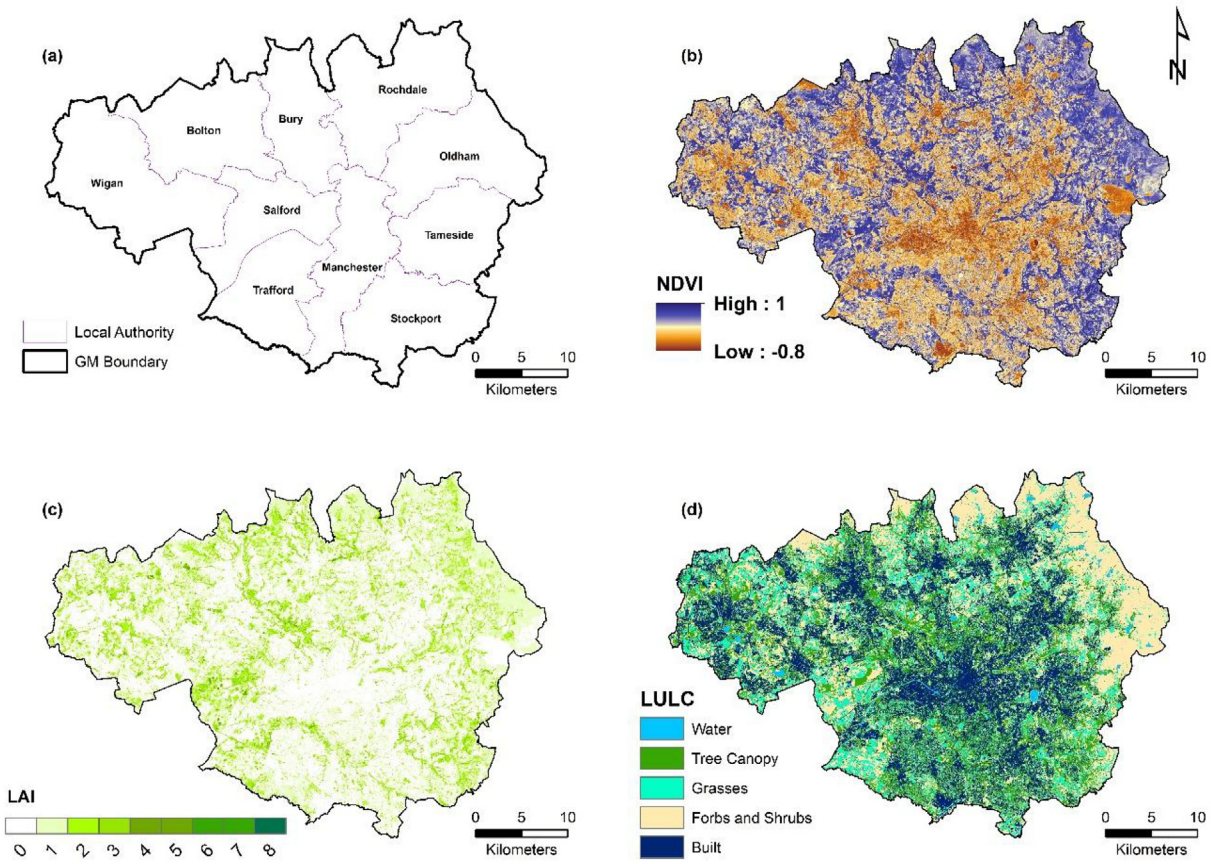


Fig. 2. (a) Greater Manchester with ten local authorities. (b) NDVI map for the whole study area. (c) LAI map, the central area is mostly zero leaf area index value. (d) Land use land cover map with five types of LULC.

produced for five buffer distances between 100 and 500 m. For LAI and LULC, similar patterns were observed. The 'exposure index' maps for LAI and LULC for the same scales are presented in Supplementary Fig. S7 and S8.

4.4. Multi-scale greenspace 'exposure index' maps

By combining the multiple 'exposure index' maps from Section 4.3, we produced multi-scale greenspace 'exposure index' maps for each metric using weights obtained from the lacunarity analysis. Table 1 lists these scales and corresponding weights. The combined multi-scale exposure mapping outputs are presented in Fig. 5. More weight was given to the larger scales (smaller buffer distance, Table 1), where landscape heterogeneity is greater and spatial characteristics were better understood. This limits the impact of aggregation. The multi-scale NDVI exposure map (Fig. 5a) therefore contains similar patterns to larger-scale exposure maps (e.g., 100 m) as it prevents the spatial characteristics of vegetation from being lost due to aggregation at smaller scales (e.g., 500 m). The combined LAI exposure map (Fig. 5b) shows high mean LAI values concentrated in specific areas within the study area, corresponding to relatively small areas of dense canopy coverage. The combined multi-scale NDVI, LAI and LULC-derived, percentage-based map (Fig. 5a,b,c) clearly highlights the densely built-up city core (Supplementary Fig. S9).

4.5. Composite multi-metric, multi-scale greenspace 'exposure index' map

Our novel composite greenspace 'exposure index' map combines all three multi-scale exposure maps (NDVI, LAI, and LULC) (Fig. 6). Compared to individual multi-scale metric maps (Fig. 5), the full composite accommodates the spatial patterns and geometric characteristics of all three metrics at varying scales. It also preserves the spatial pattern and landscape characteristics, alleviating the problem of scale effects.

4.6. Implications of the composite greenspace 'exposure index' maps

Fig. 7a illustrates that the new composite greenspace 'exposure index' map for neighbourhoods (LSOA) in the study area identifies greater differences in greenspace exposure profiles between the most and least deprived areas compared to mean NDVI, LAI and mean greenspace percentage, despite the greenspace percentage showing higher values for the least deprived deciles (IMD 8–10). The greenspace percentage metric failed to capture the variations in the most deprived deciles (IMD 1–3). Therefore, we concluded that the new composite greenspace 'exposure index' map is not only more theoretically robust than traditional methods, but when applied to the example of how exposure profiles differ by social deprivation deciles it also reveals a greater variation in overall greenspace exposure. Such variation in exposure is important in understanding overall health inequalities (Marmot, Allen, Boyce, Goldblatt, & Morrison, 2020; Mitchell & Popham, 2008).

Spearman's Rank correlation analysis of IMD, health deprivation (HealthD) and greenspace exposure using traditional metrics (LSOA aggregation) without scale effects being considered and the new multi-scale and multi-scale, multi-metric composite index data showed that there were strong significant positive correlations (Fig. 7b). However, compared to the traditional metrics without scale effects being considered the new multi-scale exposure maps and the multi-metric, multi-scale composite index illustrated greater correlation coefficients (Fig. 7b, correlations with deprivation). These correlations confirm the promise of the composite metrics for representing exposure gradients.

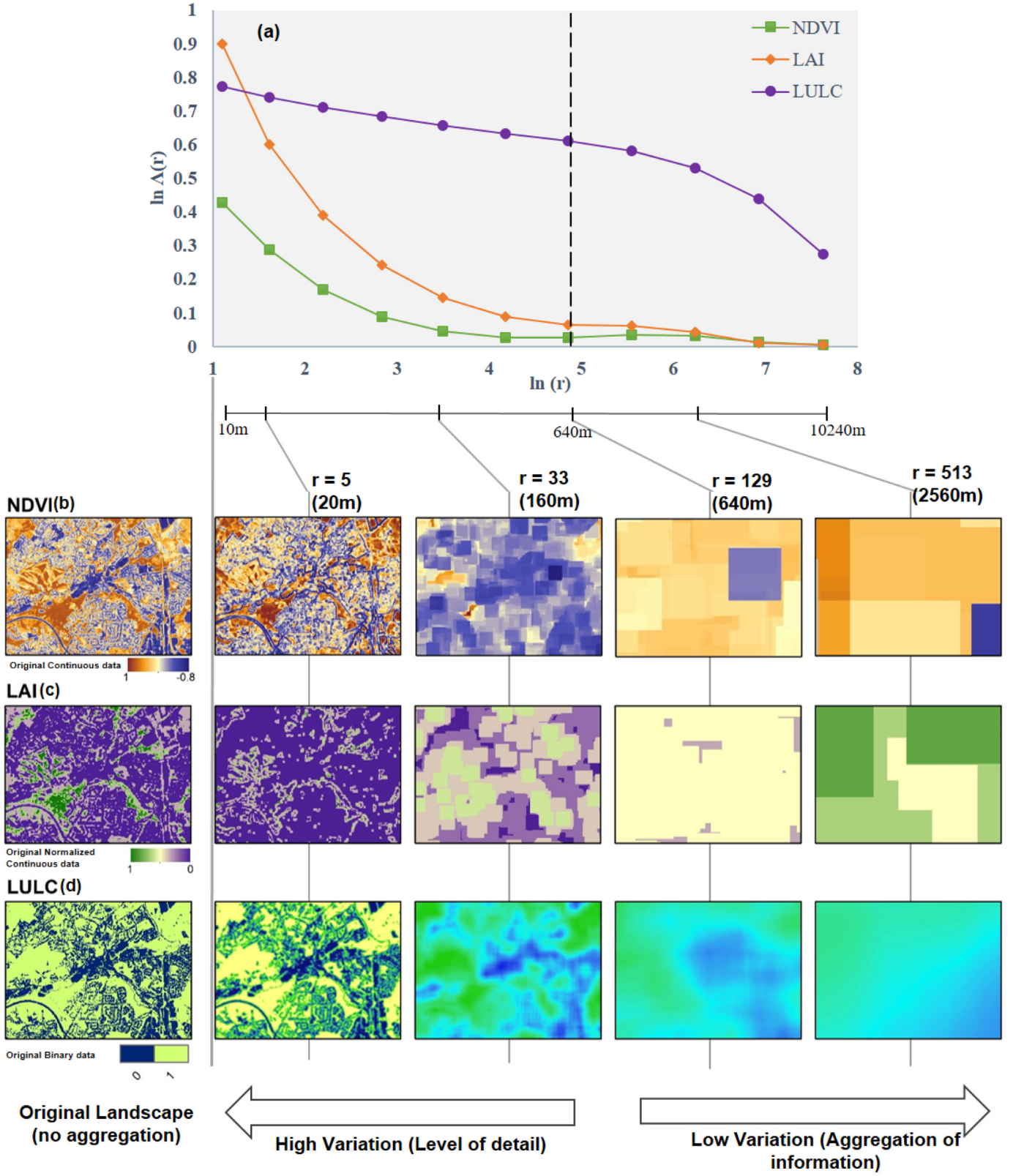


Fig. 3. (a) Lacunarity curves comprising *log* lacunarity plotted against *log* box size values for all three greenspace metrics. (b) Comparisons of box mass (range) values for NDVI for example box sizes for a subset of the full study area. (c) Comparisons of box mass (range) values for normalised LAI for example box sizes. (d) Comparisons of box mass (sum) values for LULC for example box sizes.

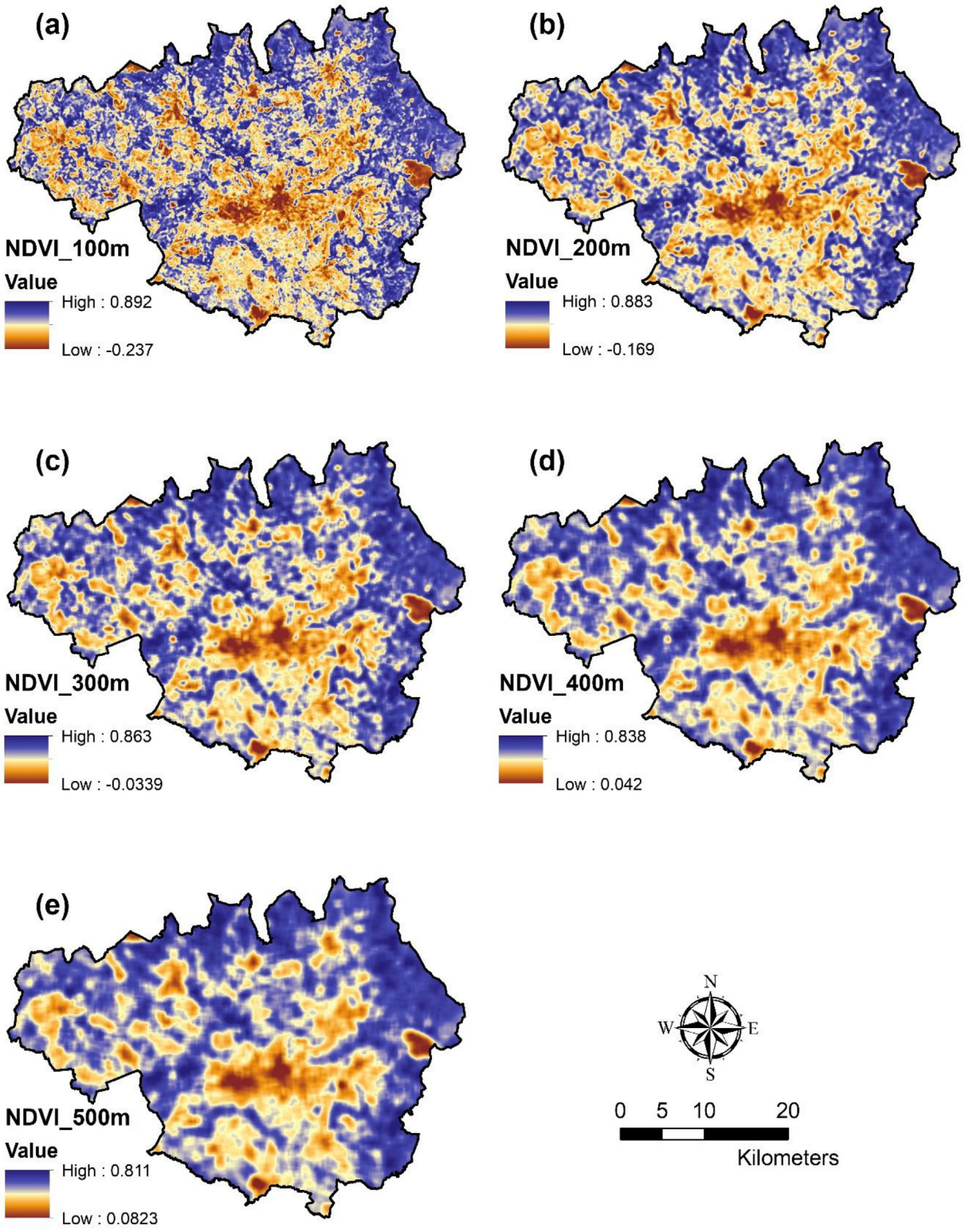


Fig. 4. NDVI maps using box sizes of (a) 100 m; (b) 200 m; (c) 300 m; (d) 400 m; and (e) 500 m.

5. Discussion

5.1. Scale effects in greenspace exposure

Our study is the first to investigate the scale sensitivity of greenspace exposure metrics using the systematic approach offered by lacunarity analysis. This study also advanced the state-of-the-art by providing a better understanding of the scale effects in representations of greenspace exposure at different buffer sizes. The existence of MAUP in satellite imagery based NDVI data has been suggested previously (e.g. Reid et al., 2018; Su et al., 2019). However, we provide new insights into the specific effects in commonly used urban greenspace metrics and present a transferable methodology through which scale effects can be analysed and handled. All three greenspace metrics

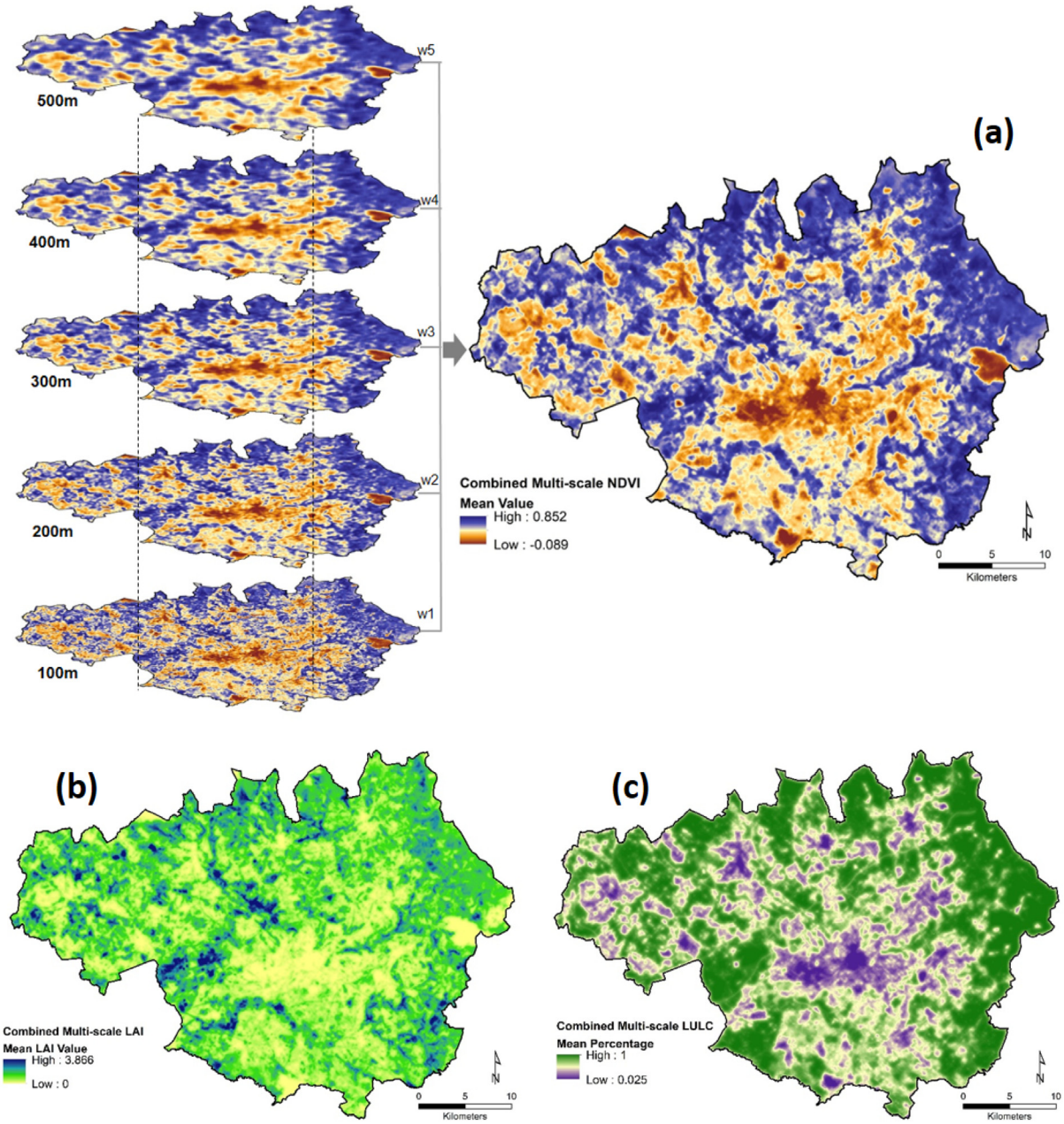


Fig. 5. The five scales and related weights combined to produce a multi-scale greenspace 'exposure index' map for Greater Manchester for each metric (a) NDVI, (b) LAI, (c) LULC (percentage of green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

considered in our study confirmed susceptibility to scale effects and MAUP when different spatial scales and areal units of measurement were applied (Fig. 3). Furthermore, the type of metrics (e.g., LAI or NDVI) and data (e.g., Sentinel-2, Landsat-8 images) utilised and their geometric patterns were also sensitive to different degrees of variance depending on the scale that was applied.

The lacunarity analysis results in the present study show the upper limit for scale or buffer distance values (~640 m for Sentinel-2, ~480 m for Landsat-8) above which real-world landscape characteristics and image textures might no longer exhibit variability because of a reduction in variance caused by aggregation of the landscape patterns (see Fig. 3b, c, d). Malhi and Román-Cuesta (2008) and Myint and Lam (2005) identified similar patterns for vegetation while using satellite images. Fotheringham and Wong (1991) and Tate (2001) demonstrated that such phenomena can produce uncertainty in the statistical analysis of relationships with other variables. Applying buffer distances greater than this upper limit affects the variance and normality of the data distribution, and so influences the results of statistical analyses (e.g., regression). A worked-example of the effects of increasing buffer distances on variance and normality of the greenspace exposure measurement has been conducted for the case study area (see *Supplementary note-3)*. These findings are critical, as the reduction in variance in resultant greenspace exposure values may ultimately influence the statistical relationships with health data, which relies on the variance and covariance among these data and their distribution. Therefore, selecting the appropriate buffer distances below the upper limit of the acceptable range of scales should be a key consideration.

Considering the inconsistency observed in effect sizes and significance with varying buffer sizes applied to set study scales (James et al., 2014; Labib et al., 2020; Su et al., 2019; Lee et al., 2019), future studies should consider investigating and reporting the variance of greenspace exposure metrics at different buffer distances before

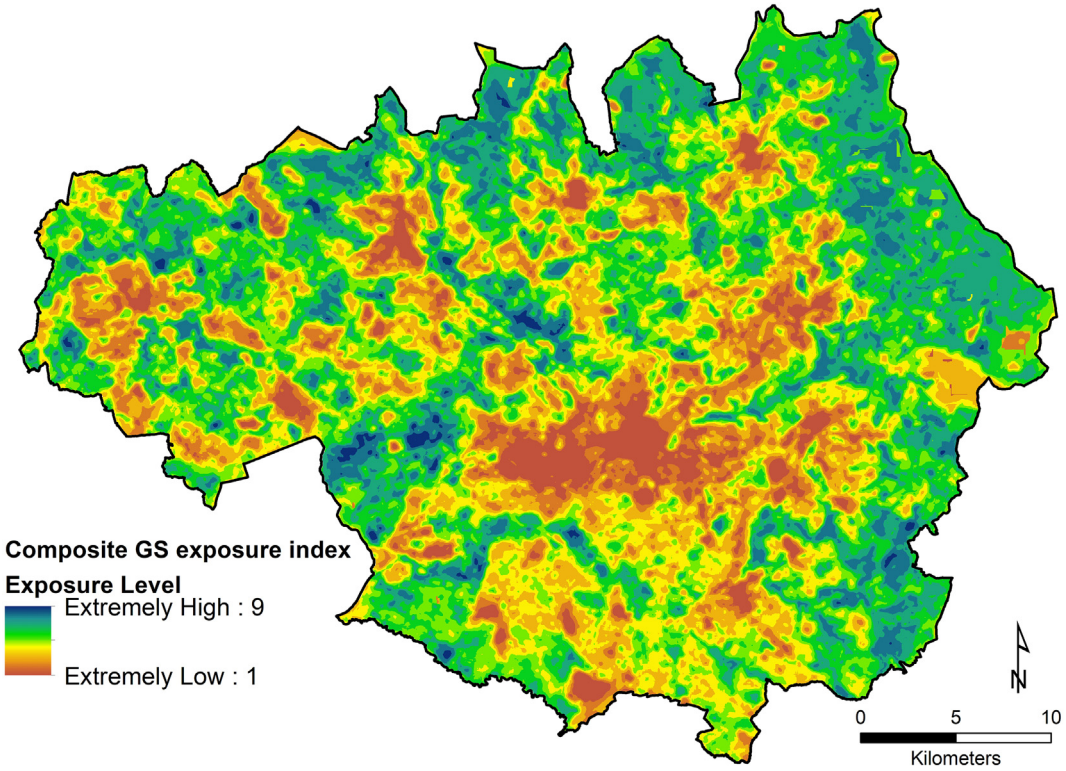


Fig. 6. Composite multi-metric, multi-scale greenspace 'exposure index' map.

conducting any statistical analysis (e.g., regression). Care should be taken in using larger buffers, as over-aggregation at larger buffer distances tends to reduce the variance and result in inconsistent associations with other variables in the models due to the presence of scale effects related to MAUP (Reid et al., 2018).

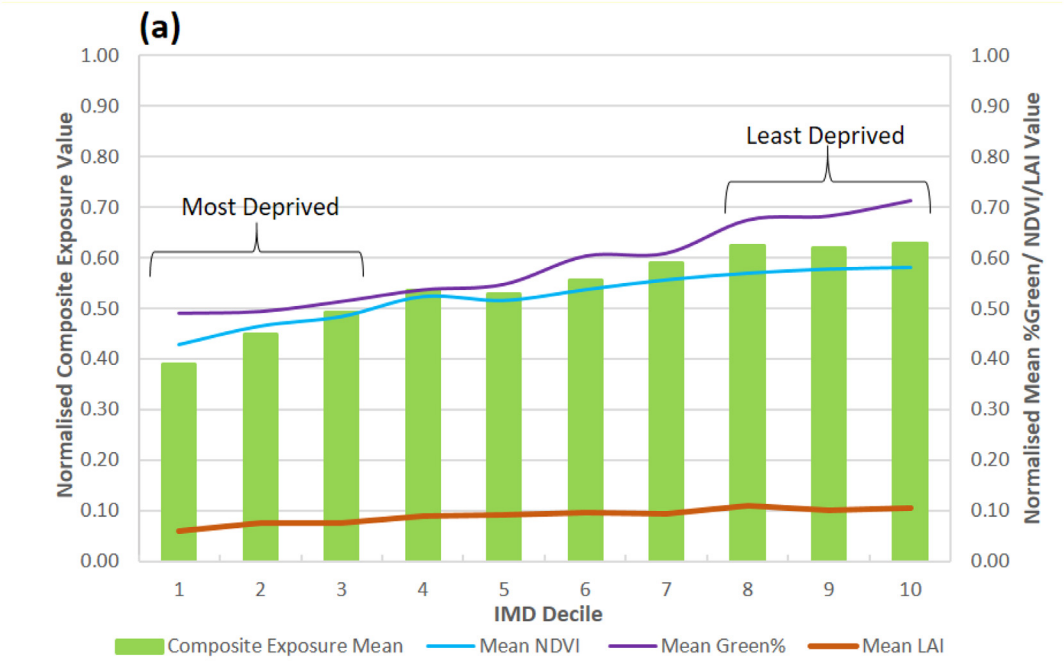
5.2. Composite greenspace 'exposure index' mapping

Previously Su et al. (2019) and Browning and Lee (2017) have proposed several approaches to counter scale effects. Their recommendations included utilising activity space to determine the exposure area (with GPS data), investigating the sensitivity of different buffer distances when determining the scale, and using nested (doughnut-shaped) buffers rather than circular buffers. However, using activity spaces for determining buffers remains quite subjective and prone to error as it depends on user mobility patterns, and the method is vulnerable to selective mobility bias. Furthermore, the approach is also highly resource-intensive (Chaix et al., 2013; Helbich, 2018). Kuo, Browning, Sachdeva, Lee, and Westphal (2018)and Browning and Locke (2020) suggested using a nested buffering approach; however, Zhang et al. (2018) indicated that because nested buffers are a subset of circular buffers, such approaches produce weak associations between greenspace and health indicators. Considering all the above arguments, our approach provides a more comprehensive and less resource-intensive solution to addressing scale effects in the analysis of satellite data for the purposes of greenspace-based studies.

We produced maps for five equal interval buffer distances (i.e., maps with box sizes of 100, 200, 300, 400, and 500 m). We then produced a combined multi-scale exposure map for each metric by weighting the different box size maps using weights derived from the lacunarity analysis. These multi-scale single metric 'exposure index' maps were produced below the upper limit of scale variability. They provide opportunities to select one or more scales based on the pathways of health benefits under consideration (Labib et al., 2020; Markevych et al., 2017; Su et al., 2019). These maps facilitate city-wide dynamic greenspace exposure assessment, as discussed by Helbich (2019b, 2018), Zhang et al. (2018); Park and Kwan (2017). Furthermore, because these maps assess and represent the potential exposure of all locations (cells) across the area of interest it means that greenspace exposure profiles can be readily generated for the different locations where the subject lives, works, or travels by incorporating the data such as GPS data tracks, participatory mapping data or spatial diary entries.

We have also developed a composite multi-scale, multi-metric greenspace 'exposure index' map (Fig. 6). Our new index has several benefits over the single multi-scale metrics. Firstly, it is scale-weighted (as above) and so is not as susceptible to scale effects. Secondly, our map includes multiple metrics (i.e., NDVI, LAI, and LULC), which account for different vegetation characteristics (e.g., NDVI and LAI indicate vegetation density and volume, whereas percentage greenspace indicates the overall presence or absence of greenspace; James et al., 2015). In combination, these metrics provided a comprehensive measurement of potential greenspace exposure across the study area. Consideration of both multiple scales of analysis and metrics supports the analysis of multifunctional benefits of greenspace (Labib, 2019), and reduces uncertainty regarding the geographic context problem noted by Kwan (2012). Thirdly, our index standardises the greenspace exposure level, meaning that values can be directly compared across multiple study areas (unlike approaches such as interquartile or quantile ranges that are common in the literature). Finally, the overall composite map can be used identify greenspace exposure levels for fixed points (e.g., home, office), for neighbourhoods (areas), and along the paths (lines) people travel. This eliminates the need to use arbitrary buffer distances to measure exposure at varying locations as each pixel has the value for multi-scale, composite exposure measurement.

We demonstrated an example of neighbourhood-based exposure profiling for LSOAs in Greater Manchester. Our example illustrated that the new exposure maps can be used to study the spatial variation in exposure in different neighbourhoods and our multi-scale as well as multi-metric, multi-scale composite maps both showed stronger correlations with IMD and health deprivation than traditional non-scaled



(b)

	LSOA Aggregation			Multiscale			Composite GS Index
	LULC	NDVI	LAI	LULC	NDVI	LAI	
IMD	0.371	0.344	0.28	0.416	0.378	0.306	0.384
Health D	0.363	0.34	0.28	0.402	0.371	0.307	0.375

All correlations are significant at the 0.01 level (2-tailed).

Fig. 7. (a) Comparisons of the Index of Multiple Deprivation (IMD) with traditional greenspace metrics, and the new outputs for urban neighbourhoods (LSOA), (b) correlations aggregated at neighbourhood scale (i.e. LSOA), new multi-scale metrics, and the overall multi-scale, multi-metric composite index.

exposure metrics (see Section 4.6). These initial results are important because they imply that traditional methods may underestimate the differences in exposures which exist between urban communities, and that inequalities in beneficial exposures may in fact be larger than is frequently reported. Of particular note is the markedly lower exposure to greenspace seen in the most socially deprived areas (Category 1, Fig. 7) in the new multi-scale index maps compared to maps using traditional non-scaled metrics. We therefore conclude that our novel multi-scale exposure maps and composite multi-metric, multi-scale greenspace 'exposure index' map have the potential to reduce errors in greenspace exposure estimation.

5.3. Strengths and limitations of this study

Our multi-scale and composite index methods and maps provide opportunities to overcome scale issues in the use of moderate resolution satellite images for greenspace exposure assessment in typical urban contexts. These maps can be used as a standardised method for both static and dynamic exposure assessment and are easy to understand for non-experts.

There are several areas in which this study could be further developed. For example, we have only included NDVI, LAI and LULC metrics, meaning that our results do not account for exposure to blue spaces. The addition of suitable measures such as the Normalised Difference Water Index into the analysis would remedy this. Secondly, because of limitations in computational resource and time, we have only tested the methods on moderate-coarse resolution satellite images (Sentinel-2, Landsat-8). Future research should include the application of this method to high-resolution satellite images such as GeoEye-1 or WorldView-3. Based on our experimental observations, we would expect that the upper limits of scale variance would be slightly larger for higher resolution imagery, though this requires further investigation. Thirdly, we only investigated scale effects in satellite image based exposure metrics as a measure of greenspace abundance. Recent work has utilised street view images to measure how people actually observe greenspace (Helbich et al., 2019; Li & Ghosh, 2018; Lu, 2019; Wang et al., 2019). Scaling effects in observational metrics (e.g., derived from street view images) should therefore also be explored in future studies. Fourth, we have only presented an analysis from a single case study region. Further research should examine across several different cities in several different parts of the world in order to better understand the variation in scale sensitivity that arises from the spatial configuration of the study region. Finally, we did not test our new exposure index with explicit health indicators to test its robustness. Instead, we compared with IMD, and health deprivation, which are pseudo indicators for health and wellbeing (Cebrecos et al., 2018). Future research might include an analysis of various explicit health indicators in order to use the benefits of this new methodology in order to better understand the relationship between greenspace and health and wellbeing.

6. Conclusion

The main focus of this study was to investigate scale sensitivity and scaling effects in greenspace exposure assessments. The results of our

lacunarity analysis indicate that the greenspace metrics commonly used in recent studies are scale sensitive depending on the buffer distances used to set the scale of analysis. Based upon this analysis, we identified the range of buffer distances within which a change in scale impacts upon analysis (beyond ~640 m for Sentinel-2 and ~ 480 m for Landsat images). Beyond the upper limit of the acceptable range of scales, the characteristics of urban greenness are no longer properly captured as they tend to be over-aggregated. In order to address observed scale effects, we produced multi-scale greenspace 'exposure index' maps for a large study area, based upon a range of scales from within this range. These maps can be used to measure greenspace exposure at varying locations depending on the pathways relevant to the specific scale. Second, we developed a composite multi-scale greenspace 'exposure index' map that combines multiple metrics and scales. The multi-scale and composite exposure metrics improve the basis for greenspace exposure assessment by reducing scaling effects and uncertainty when selecting greenspace metrics for different health pathways of greenspace exposure benefits. We then illustrated the benefits of this improved model by using it to examine the relationship between greenspace and deprivation, which revealed a greater difference in potential exposure to greenspace between the most and least deprived in the study region, a pattern that has wider implications in understanding wider health inequalities (Marmot et al., 2020). Our map outputs provide an efficient and scale-sensitive means to estimate greenspace exposures for points (e.g., residential addresses), lines (e.g., walking routes), and areas (e.g., neighbourhood) exposure units, permitting a better characterisation of exposure profiles compared to traditional exposure assessment approaches. The method is readily transferable to the analysis of other environmental 'landscapes' associated with exposure such as air pollution and high temperatures (e.g. LST) and thus may have significance beyond the study of greenspace and health associations.

Declaration of Competing Interest

The Authors declare no conflict of interests.

Acknowledgements

The authors would like to thank the anonymous reviewers and editors of this paper for their constructive comments and suggestions. S.M. Labib is funded by the School of Environment, Education and Development Postgraduate Research Scholarship at the University of Manchester. Elements of this work were carried out as part of the Green Infrastructure and the Health and Well-Being Influences on an Ageing Population (GHIA) project (2016–2019) http://www.ghia.org.uk. This project is funded by the Natural Environment Research Council, the Arts and Humanities Research Council and the Economic and Social Research Council under the Valuing Nature Programme. NERC grant reference number NE/N013530/1.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compenvurbsys.2020.101501.

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